Alex Cason

Curriculum Vitae

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GitHub: Physics-is-awesome · YouTube: @Eudaimonia_Society
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EDUCATION

University of Tennessee, Knoxville — Expected May 2030
B.S. in Physics and Mathematics (double major) — Knoxville, TN

RESEARCH EXPERIENCE

Assistant Researcher — August 2024 – Present
University of Tennessee, Knoxville – Dr. Mezzacappa — Knoxville, TN

- Research gravitational-wave emission from core-collapse supernovae using simulation data produced with the Flash-X code.
- Wrote and maintain Fortran code that reads core-collapse supernova simulation output and generates gravitational-wave data for analysis.
- Analyze the resulting gravitational-wave data to characterize the signal produced by the simulated explosions.
- Currently developing tools to observe core-collapse supernova gravitational-wave memory in next-generation detectors.
- Using Supercomputer UT-K ISAAC

Nuclear Electric Propulsion Thruster Design Subteam Lead — August 2026 – Present
SSTA

- Lead the subteam, of seven members, in designing a nuclear electric propulsion (NEP) thruster, setting project goals and technical direction from knowledge of numerical plasma software.
- Schedule and run weekly technical meetings.
- Teach team members the fundamentals of plasma physics, numerical methods, and scientific coding.

Contributor — November 2024 – Present
PlasmaPy (open-source plasma physics package)

- Enhance computational tools and simulation frameworks used in plasma physics research.
- Participate in scientific discussion and perform code review on community contributions.
- Develop educational tools for the plasma physics community.

Contributor — October 2026 – Present
GeometricIntegrators.jl (open-source geometric integrators library)

- Currently working on developing Developer's Documentation
- Initial stages of developing Hamilton-Pontryagin-Galerkin Integrators

SELECTED RESEARCH AND SOFTWARE PROJECTS

Independent computational projects; most are documented with derivations and code walkthroughs on my YouTube channel.

Relativity, gravitation, and Hamiltonian mechanics

- **BSSNOK Code.** Numerical-relativity code that evolves spacetime using the BSSNOK formulation of Einstein's equations, currently developed for short-term simulations of kerr black holes.
- **Planck force simulation.** Simulation of the Planck force using the Einstein Toolkit.
- **Relativistic rocket equation (Hamiltonian).** Hamiltonian formulation of relativistic rocket dynamics.

- **WHCKL celestial mechanics simulation.** Extremely optimized Celestial-mechanics simulation. Able to simulate two billion years of fifteen bodies in the solar system with an energy drift of about 10^{-5}
- **Hamiltonian framework for the ICF implosion rocket equation.** Model based on Rayleigh–Taylor instabilities.

Plasma physics and magnetohydrodynamics

- **MHD model in Fortran.** Basic Fortran implementation of magnetohydrodynamic equations.
- **Hamiltonian MHD and variational MHD.** Geometric formulations of MHD, including variational integrators for reduced MHD (RMHD).
- **Metriplectic plasma simulation.** Simulation of plasma dynamics combining Hamiltonian (conservative/symplectic) and dissipative structure(Riemannian).
- **MPD rocket simulation.** Basic MHD-based rocket simulation of MPD thruster.
- **3D compressible plasma using lattice Boltzmann methods.** Numerical methods for compressible plasma flow.

Astrophysics and computational methods

- **DG-SG Fluid.** Discontinuous Galerkin simulation of self-gravitating fluid. Enforced Positivity and energy conservation
- **Simulated annealing.** Optimization models using simulated annealing.
- **PPO agent.** Proximal Policy Optimization (reinforcement learning) agent trained to play an Asteroids-style game.

TEACHING, MENTORING, AND OUTREACH

Content Creator, YouTube (`@Eudaimonia_society`) May 2026 – Present
Educational physics channel

- Produce advanced educational videos on physics and scientific computing, with accompanying code.
- Topics: Derivation and explanation of above software project, introductions to methods in structure preserving integrators, and derivations of advanced topics in mathematical physics.
- Derivations covered: the Navier–Stokes equations from the Boltzmann equation and the Boltzmann equation from Hamiltonian dynamics, motivation of Metriplectic Dynamics, and Lie Operator splitting.

Founder August 2024 – May 2026
Scientific Visionaries of America

- Founded and promoted a high school science club; taught classes on advanced subjects and led activities including group discussions, games, experiments, and public speaking.
- Organized the inaugural Wilson County Science Fair.
- Prepared members for JSHS, Science Olympiad, Science Bowl, and the Science and Engineering Fair.
- Helped member win interdisciplinary award in MTSEF

Assistant Instructor, Karate August 2021 – May 2026

- Led warm-ups and workouts, taught students, and corrected their technique.
- Worked with over twenty students
- Taught classes with age ranges from eight to forty

LEADERSHIP EXPERIENCE

Chapter Chief and Fall Fellowship Director October 2024 – May 2026
Order of the Arrow

- Led chapter operations: scheduled and ran member and leadership meetings, and managed the Secretary, Treasurer, Unit Elections Chairman, and Vice Chief.
- Set and completed chapter goals; planned events, campouts, and community service projects.
- Organized OA elections and the district camporee.
- Planned and directed Fall Fellowship, leading 30+ members to host over 211 attendees.

- Staffed Klondike, Winter Camp, Inductions, and Fall Fellowship, serving as key staff for the Fall Camporee and Fall Fellowship.
- Taught at Scouting University.
- Choicen to teach at National Youth Leadership Training

S1 Officer and PT Officer August 2024 – May 2026
Junior Reserve Officers' Training Corps (JROTC)

- Managed unit paperwork and tracked attendance, service hours, and physical training scores.
- Planned physical training sessions.

Senior Patrol Leader August 2021 – 2022
Boy Scouts of America

- Scheduled and led troop meetings.
- Planned and led troop camp-outs and activities.

TECHNICAL SKILLS

Programming (experienced): Fortran, Python, LaTeX
Programming (comfortable): Julia, Bash, Lisp, OpenMP
Programming (basic): C++, C
Simulation and numerical software: Flash-X, OpenMHD, Gkeyll, PlasmaPy, SPIS, OpenFOAM, PLUTO, Einstein Toolkit
Environment: Linux, command-line workflows, SLURM, MPI, Paraview, VisIt

HONORS AND AWARDS

2026 Vigil Honor, Order of the Arrow
2025 Eagle Scout, Boy Scouts of America
2025 Naval Science Award at MTSEF
2025 2nd Place, Regional Oratorical
2024 Second-degree black belt, Karate

OTHER WORK EXPERIENCE

Boat Rental Attendant August 2025
Twin Creeks Marina Winchester, TN

- Parked and moved boats, cleaned and prepared them, and gave customers safety briefings.
- Prepared and stored lily pads and tubes; cleaned docks, storage bins, and electrical boxes.

COMMUNITY SERVICE

Providence Church: Assisted teaching Sunday School and leading youth sermons; served communion; helped with parking and with event set-up and take-down; brought food to the elderly and homeless; served meals at a homeless shelter; helped teach young children to read.

Order of the Arrow: Staffed Winter Camp (twice) and Klondike Camp; served food; cleaned parks and scouting reservations.

Scouting America: Served food to the elderly; cleaned scouting reservations; assisted at Eagle Scout workdays and at the American Legion tree sale.

JROTC: Assisted at the American Legion tree sale; served food to others.